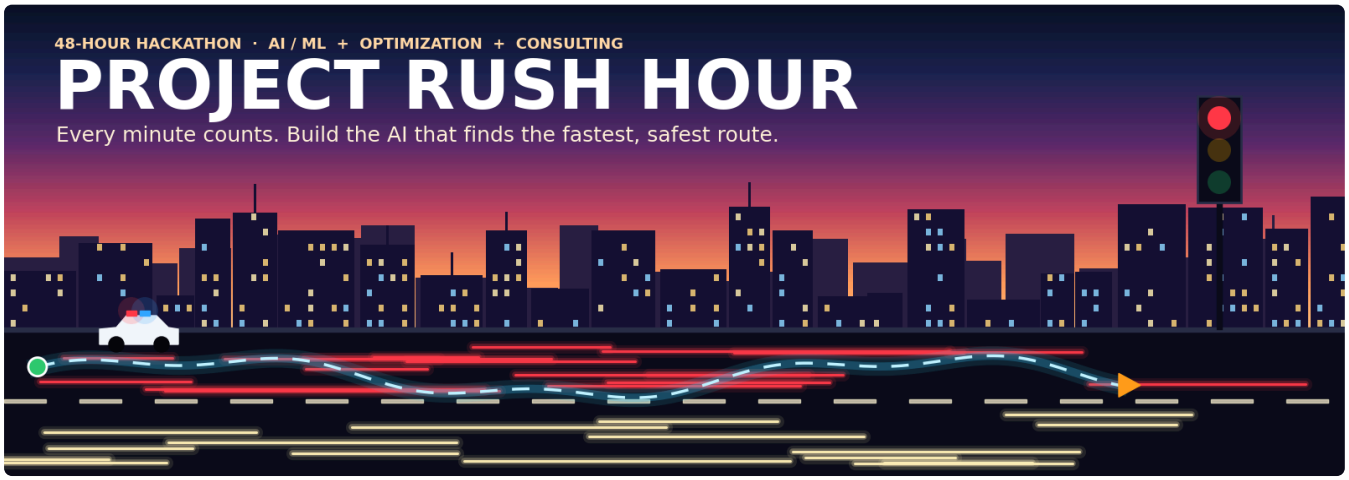
Project RUSH HOUR: AI-Powered Emergency Response Challenge

Oct 16, 2026



The mission

A large, densely populated city wants to improve its emergency response, especially in high-traffic zones where every minute of delay matters. Your team will design an AI-powered emergency response service that determines the fastest and safest path for each emergency vehicle, and make the case for how the city should set it up and run it.

Format: 48-hour hackathon · Synthetic city traffic · Python · AI/ML + graph optimization · Consulting case

Your client: the (fictional) city government. It pays for the service. Your team will deliver a complete solution: an AI routing engine, a plan for running it as a city service, and a client-ready recommendation.

> **Important:** This is a self-contained educational simulation, and the city, client, and operators are fictional. Road features, labels, and costs are synthetic. They are not real traffic observations or validated routing or safety data.

The story

Metro City is large and densely populated. In its high-traffic zones, ambulances, fire engines, and police vehicles lose precious minutes to congestion, incidents, and closed roads. The city wants to improve its existing emergency response with AI.

The proposed service works like this: a citizen dials the emergency number, the call is logged, and an AI system predicts traffic conditions and recommends the fastest and safest route. The nearest available vehicle is dispatched along that route.

The city pays for the service. The emergency vehicles are run by third-party operators, who are paid by the government for their services. Your client, the city, wants to know whether and how this service should be set up.

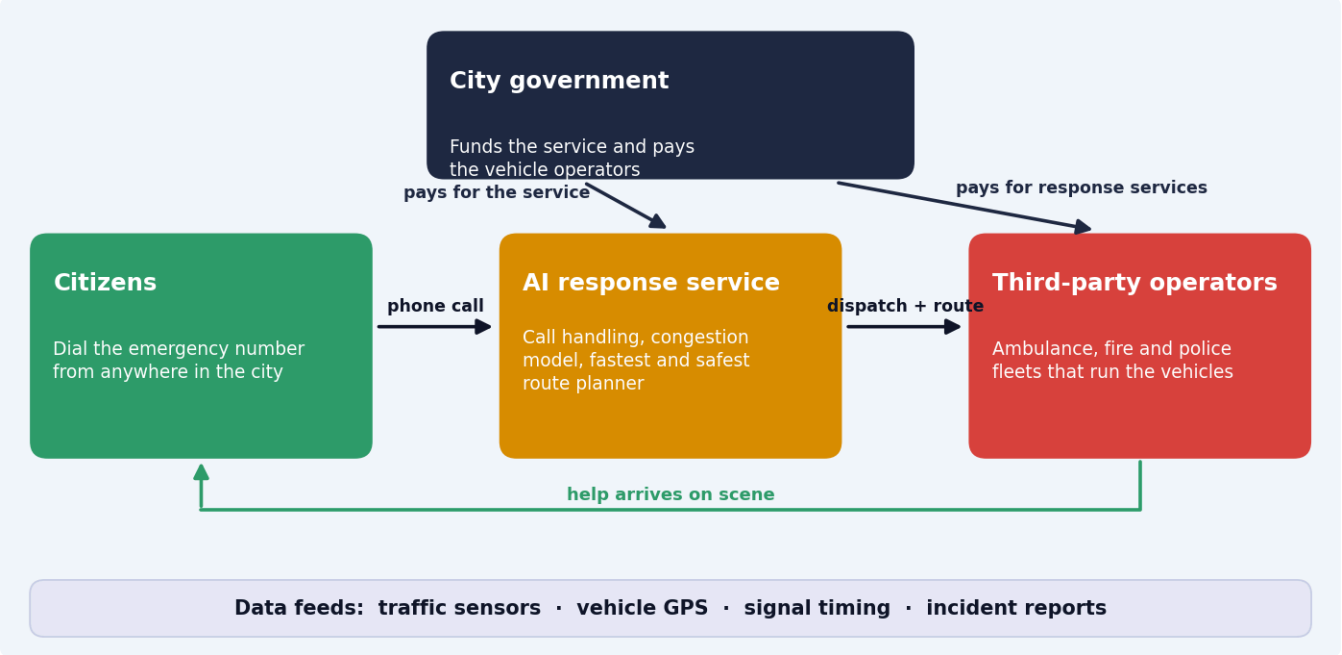
Your team plays a combined engineering and consulting team advising the city. The city wants a complete solution, and your team must answer three questions:

1. What does the traffic data tell us about how drivable each road cell is?
2. Given those predictions, what is the fastest and safest route for the vehicle?
3. How should the city set up, fund, and run this as a real emergency response service?

The shortest route is not always the fastest or safest route. A longer path may have a lower predicted travel cost. Your system should make this trade-off explicit and measurable.

The challenge

Design and recommend a complete emergency response solution for the city. It has two sides that depend on each other: an AI engine that finds the fastest and safest route for every call, and a service plan that makes it work in the real world, from the phone call to who pays for what.



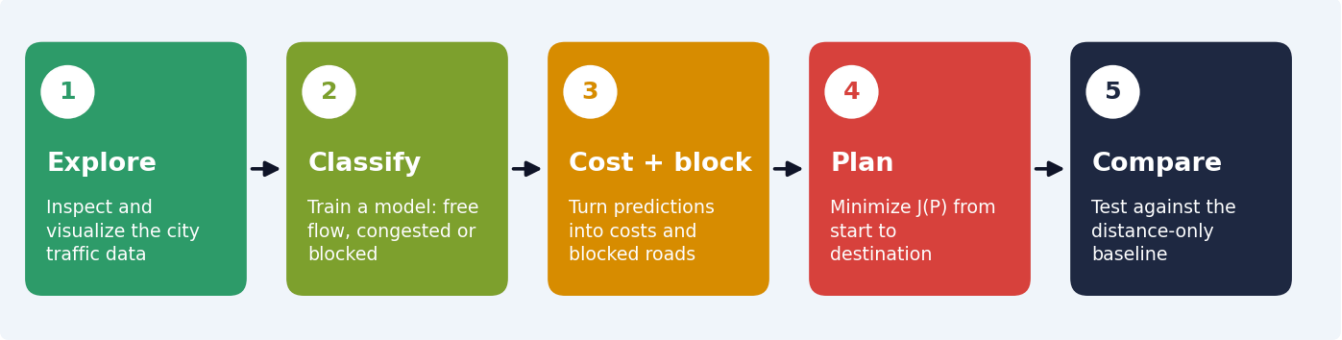
One possible service design. Your team may propose a different one.

Your solution has three parts:

1. **The AI engine.** A pipeline that predicts congestion from traffic data and plans the fastest, safest route, tested against a distance-only baseline.
2. **The service plan.** How a call becomes a dispatched vehicle, how the city funds the service and pays the third-party operators, the risks, and how to roll it out.
3. **The recommendation.** A one-page executive summary and a short deck that bring both together for the city.

Part 1: The AI engine

Build a working pipeline with five stages, from raw traffic data to an explained result.



In detail, your pipeline must:

1. Explore and visualize the supplied traffic data.
2. Train a machine-learning model to predict a cell's congestion class: free flow, congested, or blocked.
3. Use model predictions to estimate travel costs and identify cells to avoid.
4. Find a feasible route from the vehicle's start position to the incident location.
5. Compare the result with a distance-only baseline.
6. Explain the trade-off between distance and predicted travel cost.

Core machine-learning task

Predict the target column `congestion_label` from these features:

- `avg_speed_kmh`
- `signal_delay_sec`
- `traffic_density_0_1`
- `intersection_density_0_1`
- `incident_probability_0_1`
- `road_type`

Use `train_cells.csv` for training and `test_cells.csv` for held-out evaluation. The test data comes from separate synthetic city maps; do not train on the held-out test set. Suitable approaches include decision trees, random forests, gradient boosting, or another model your team can justify. Deep learning and GPUs are not required.

You may use the provided baseline and its precomputed predictions as they are. Improving the model is optional; analyzing its errors, especially missed blocked roads, is required.

Optional ML extension: predict `simulated_travel_cost` as a regression target.

Core navigation and optimization task

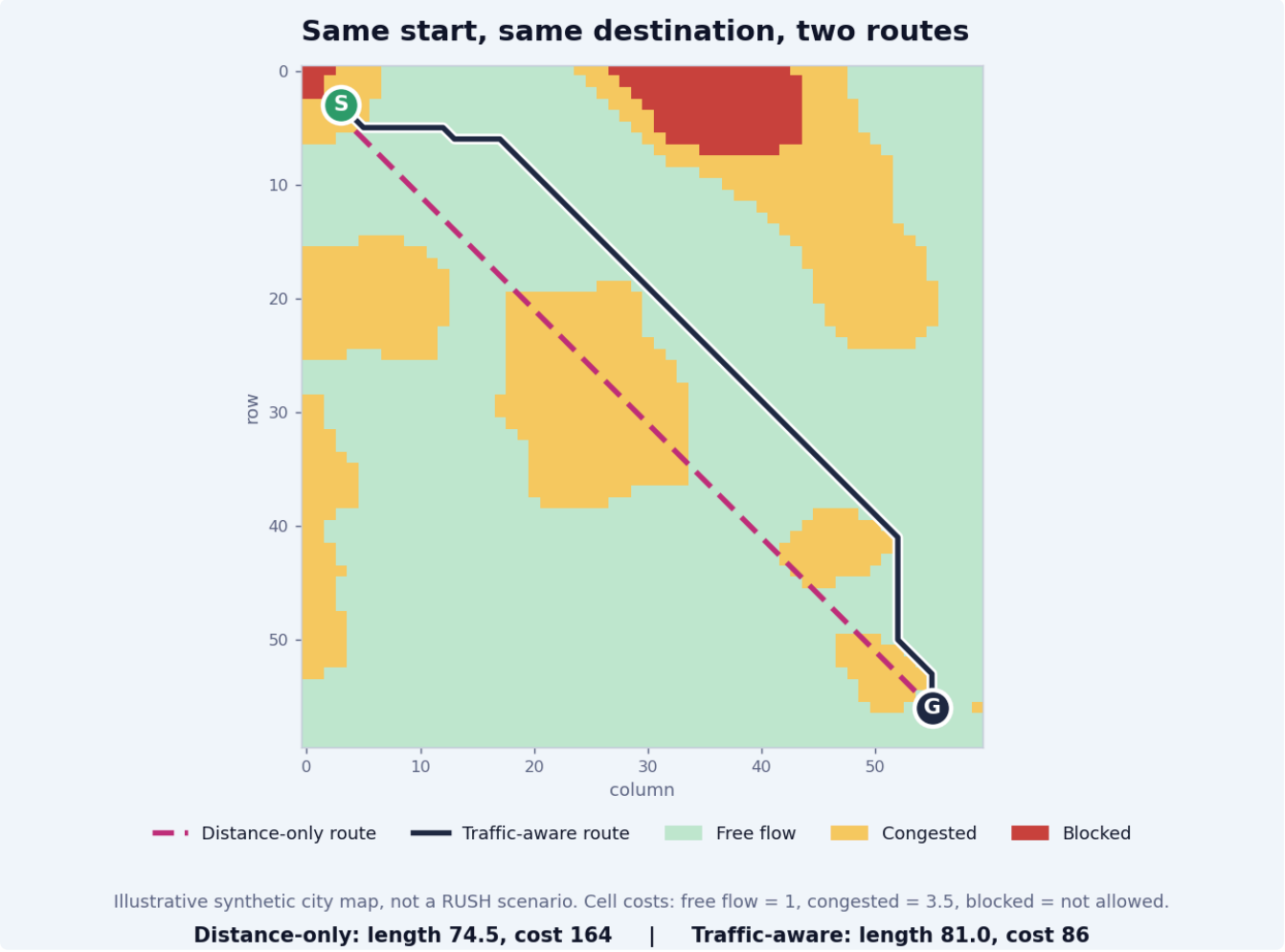
Find a route P from start S to destination G that minimizes cumulative predicted travel cost:

```
J(P) = sum over every step u -> v of:  d(u, v) × ( c(u) + c(v) ) / 2
```

Here u and v are adjacent grid cells. d(u,v) = 1 for horizontal or vertical moves and √2 for diagonal moves. c(u) and c(v) are the predicted travel costs of those cells. Think of cost as a combined measure of delay and risk: a lower cost means a faster, safer route.

- Choose and document a policy for treating cells as blocked (for example, predicted blocked labels and/or a blocked-probability threshold).
- Prevent diagonal corner-cutting between blocked orthogonal cells.
- Compare the traffic-aware route with a distance-only route that uses the same blocked-cell policy but ignores non-blocked congestion cost.
- Report both geometric route length and cumulative cost under your stated objective.

The map below shows the idea on an illustrative synthetic grid (not one of the RUSH scenarios). The traffic-aware route is about 9% longer, yet its cost is roughly half that of the distance-only route.



> **Algorithm choice:** A* is allowed and is not something teams must move away from. Dijkstra or another justified planner is also acceptable. The competition is about the quality of the complete navigation system, not about choosing a fashionable algorithm.

Part 2: The service plan

The engine only matters if the city can run it. Your client is the city government, which pays for the service. Citizens reach it by phone, and third-party operators run the vehicles and are paid by the government for their services. Cover the four core themes below; the other four are optional if time allows. All costs, figures, and assumptions are illustrative.

Theme	Priority	Questions to address
Service model	Core	How does a citizen's call become a dispatched vehicle with a recommended route? Where do call-takers or dispatchers stay in the loop, and could SMS or an app be added later?

Theme	Priority	Questions to address
Funding and contracts	Core	How does the city fund the AI service? How should it pay the third-party operators (for example, per call, per response, or with incentives tied to response-time targets)? What should the contracts and service-level agreements require? Include a rough cost estimate for a pilot.
Risks and mitigations	Core	What happens if the model misses a blocked road, a data feed fails, sensor coverage is uneven across neighborhoods, or dispatchers over-rely on the tool? Consider location-data privacy, liability, and a manual fallback. Link to your model's error analysis.
Rollout and success measures	Core	Where would you pilot first, for example one high-traffic zone? What are the go/no-go criteria, and which KPIs would you track (for example, average response time in high-traffic zones, share of calls meeting the target, dispatcher override rate, and cost per call)?
Current baseline	Optional	How does emergency response work today, and where are minutes lost in high-traffic zones? State your assumptions about current response times.
Stakeholders and accountability	Optional	What do citizens, the city, call-takers or dispatchers, and operators each need? Who makes the final routing decision, and who is accountable if something goes wrong?
Resources and cost	Optional	What data feeds, compute, call-handling capacity, integration with operators' fleet systems, staffing, and training are needed? Show a fuller cost model for pilot and full rollout, with stated assumptions.
Business impact	Optional	How does the distance versus cost trade-off translate into value for the city, such as minutes saved or risk avoided? State your assumptions and label the figures illustrative.

Part 3: The recommendation

Bring the engine and the service plan together in two deliverables: a one-page executive summary for the city (situation, options compared, recommendation, risks, and next steps) and a short deck of about 5 slides with your implementation plan, presented in a 5-minute pitch. State any business-impact assumptions, such as minutes saved, as illustrative.

Starter pack

Every team receives a student starter pack with data, baseline code, and consulting templates. Its contents, the data dictionary, and the setup steps are in the separate Project RUSH HOUR Starter Pack Guide.

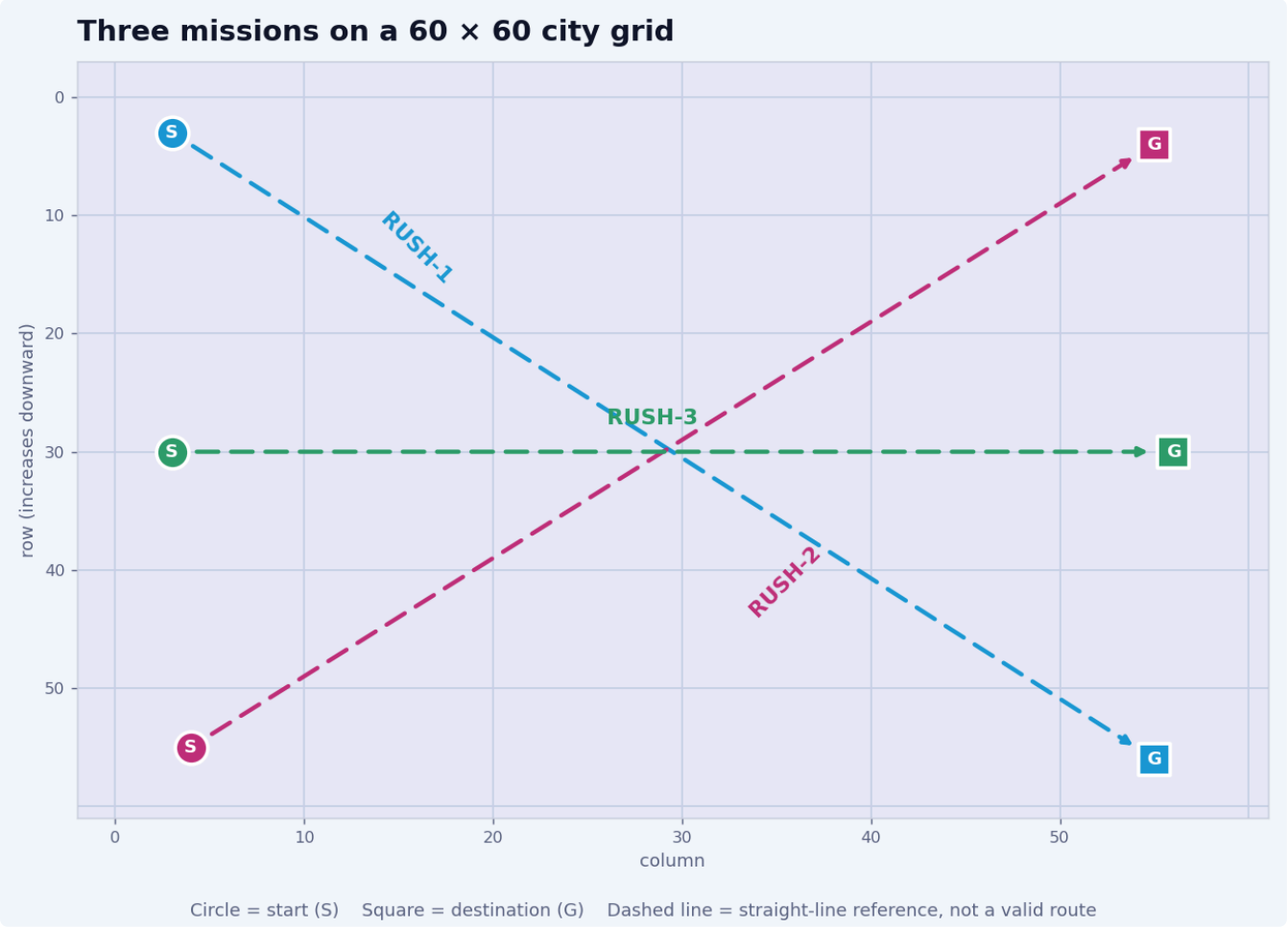
In short: the ML baseline and its predictions are provided, the cost function and blocking policy of the traffic-aware planner are left for you, and the consulting templates help you build your recommendation. Not a coder? You can still own the blocking policy, the trade-off analysis, and the business case.

Navigation scenarios

All coordinates are zero-based [row, column]. The top-left cell is [0, 0]; rows increase downward and columns increase to the right. Each map is a 60 × 60 grid.

Each scenario is an emergency call in a different part of the city. The start is where the vehicle begins, and the destination is the incident. Treat each one as a client question: your recommendation should answer it in terms the city cares about, such as response time and risk, not just route numbers.

Required: RUSH-1 and RUSH-2. **Optional:** RUSH-3, as a stress test if your team has time.



Scenario RUSH-1: The Morning Commute

Start: [3, 3] · **Destination:** [56, 55] · **Map:** `scenario_RUSH_1.csv`

- Find a feasible route using the supplied city traffic map.
- Compare the shortest-distance route with your traffic-aware route.
- Report route distance and cumulative predicted travel cost.
- Visualize both routes on the same map.

Question to discuss: How much extra distance, if any, is worth taking to reduce predicted travel cost?

Scenario RUSH-2: The Ring Road

Start: [55, 4] · **Destination:** [4, 55] · **Map:** `scenario_RUSH_2.csv`

- Evaluate whether your model and policy generalize to a different city layout.
- Find a route that avoids cells your policy classifies as blocked.
- Explain which traffic features most influence the chosen route.
- Compare the results with RUSH-1.

Question to discuss: Does the same model and cost policy work equally well across different city layouts?

Scenario RUSH-3: The Emergency Run (optional stress test)

Start: [30, 3] · **Destination:** [30, 56] · **Map:** `scenario_RUSH_3.csv`

- Replan from the new start/destination pair.
- Compare distance-only and traffic-aware routes.
- Try a more conservative blocking policy, such as blocking cells whose predicted blocked probability exceeds a threshold you justify.
- Explain how and why the route changes.

Question to discuss: How should a vehicle balance caution against the risk of taking a much longer route?

> **Scenario design:** RUSH-1 and RUSH-3 intentionally share the same underlying city map but have different start/destination pairs. This tests whether the planner responds to changed mission geometry rather than merely memorizing a route.

Evaluation and judging

Projects are judged on both by technical depth (about 50%) and consulting impact (about 50%). Organizers may adjust weights before the event.

Category	Suggested weight	What judges look for
Route quality	15%	Feasible path, lower stated predicted cost versus baseline, fair comparison, correct movement constraints
ML quality and error analysis	10%	Held-out metrics, macro F1, confusion matrix, thoughtful error analysis, especially missed blocked roads
Robustness and experimentation	10%	Multiple scenarios, threshold experiments, failure analysis
Visualization, explanation and reproducibility	15%	Clear traffic display and routes, readable metrics, clear cost function, justified decisions, repeatable results

Category	Suggested weight	What judges look for
Client recommendation	15%	Clear framing of the city's problem, options compared, a well-supported recommendation, next steps
Implementation plan	15%	Service model, funding and contracts with third-party operators, resources and illustrative cost, rollout plan and KPIs
Risk analysis	10%	Risks identified and mitigated, limitations stated, ties to the model's error analysis
Business insight and pitch	10%	Trade-offs translated into value for the city under stated assumptions, structured storytelling, confident answers to questions

A longer route can still be better if your team demonstrates a meaningful reduction in cumulative cost under a clearly stated objective. Teams must report the baseline and metrics; unsupported improvement claims should not receive credit.

Minimum viable submission

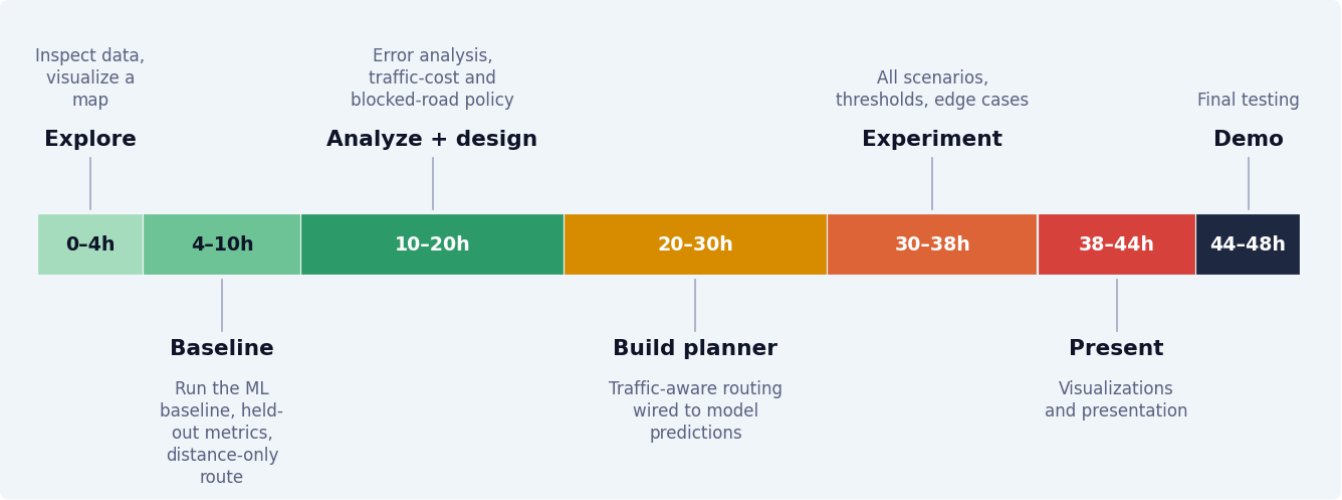
- [ ] Reproducible source code and run instructions
- [ ] A trained or justified ML model, held-out results, and at least one error analysis, including missed blocked roads
- [ ] A working traffic-aware route planner, compared with the distance-only baseline using stated metrics
- [ ] A visualization of at least one scenario
- [ ] A short explanation of the cost function, blocking policy, trade-offs, and limitations
- [ ] A one-page executive summary for the city: situation, options compared, recommendation, risks, and next steps
- [ ] A short deck (about 5 slides) with your implementation plan (service model, funding and operator payment approach, risks, rollout and KPIs), presented in a 5-minute pitch that states any business-impact assumptions as illustrative

Optional extensions

These are optional and are not required for a successful submission.

- Scenario RUSH-3 as a stress test
- Regression for numerical travel cost
- Uncertainty-aware planning
- Comparison of multiple classifiers
- A limited fuel or battery budget
- A priority reward for reaching a critical destination
- Interactive start/destination selection
- A comparison of two or more service models or operator payment models, with a simple cost-benefit view
- A detailed cost model and a phased rollout roadmap with go/no-go criteria

Suggested 48-hour workflow



Teams can work on ML and route planning in parallel. A working, measurable baseline is more valuable than an unfinished sophisticated model.

The consulting work runs in parallel too: frame the client problem early, and use the 38–44h block to finish your executive summary and short deck.

What success looks like

At the final demo, a judge should be able to choose a scenario and see:

- The city grid and predicted congestion
- The distance-only route
- Your team's traffic-aware route
- The geometric distance and cumulative predicted cost of each route
- A clear explanation of the trade-off, risk policy, and limitations

- A client-ready recommendation with its service model, funding and operator payment approach, illustrative cost, risks, rollout plan, and next steps

The goal is not to build a real city-scale navigation system in 48 hours. It is to build and evaluate a small, intelligent decision-making system that combines machine learning with path optimization.

Your code, executive summary, and deck make strong portfolio pieces for both consulting and tech applications.

Good luck, drivers.